

What is the most (i) time-efficient and (ii) space-efficient data structure in your opinion for satisfying the said application? Justify your answer.

- (b) Cite an example where the use of B⁺-tree for storing the database is disadvantageous?
- (c) Compare the performances of time taken by a heap memory and an externally sorted file to perform scanning on a big relation stored therein.

(3+3)+2+2

4. (a) Suppose the COVID-19 vaccination details of all the Indian citizens are stored in a database. The database comprises the following two tables with the schema defined therein. The keys corresponding to each table are underlined.

CITIZENS = <aadhaar: integer, *name*: string, *state*: string>

VACCINATION = <vid: integer, *aadhaar*: integer, *dose*: integer, *date*: date>

Write the following queries in SQL.

- (i) Find the names of all the vaccinated citizens who belong to multiple states. Note that, state is a multi-valued attribute.
 - (ii) Find the names of all the citizens who are not yet vaccinated. Note that, not all the citizens are vaccinated.
 - (iii) Find the aadhaar numbers of all the citizens who are double vaccinated by the end of 2021. Note that, the date format is DD-MM-YYYY.
 - (iv) Find from which state the first vaccination was done. Note that, the vaccination id is an auto-incremented integer that starts from 1 and increases by 1 in the next entry.
- (b) Suppose a table, having a key attribute and a descriptive attribute, is to be created using SQL. The descriptive attribute present in the table stores horoscope details of the people in a country with billions of entries. Suggest an appropriate (in terms of time/space efficiency) data type for the said attribute. Justify your answer.

(2+2+2+2)+2